S.U.E. Mini (v2)
(Sound Unit Expansion)
MK-XI Instructions

Crash Works 3D
Web: https://www.crashworks3d.com
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Overview

The S.U.E board is a Sound Expansion Unit that easily connects to the A.L.I.S.H.A board to add sound effects to your Iron Man helmet, suit, or Cosplay prop. It's "plug and play" feature allows the user to easily connect the two boards together with existing ports located on A.L.I.S.H.A. and S.U.E..

Specifications

The S.U.E board has a DFPlayerMini module (mp3 player), SD card slot, on-board amp, and auxiliary audio port. For audio output you have the choice of using the on-board amplifier connected to a standard 8-ohm (1 ~ 3 watt max) speaker, which will operate in mono sound. Alternatively you can use a speaker (i.e. bluetooth portable speaker, or powered speaker) connected via the aux port.

The sounds currently used for the Iron Man helmet are available for download from the Crash Works 3D github https://github.com/crashworks3d/Iron_Man_Servo . These sound files need to be added to your SD card.

SD Card Formatting & Adding Sound Files

Please visit our Github to download the PDF guide for SD Card Formatting & Adding Sound files: https://github.com/crashworks3d/Iron_Man_Servo/blob/master/Crash Works 3D - S.U.E. SD Card Formatting Instructions v1.0.pdf

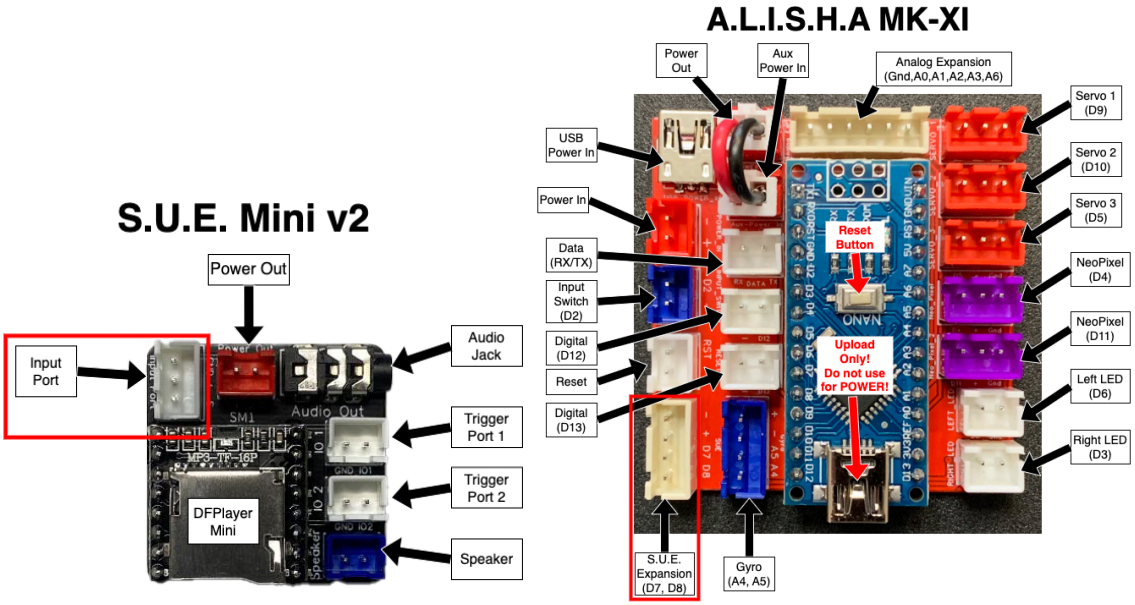
Updating the MK-XI Code

In order to enable the S.U.E. board you will need to update the code on the MK-XI. The code for the MK-XI can be found here: https://github.com/crashworks3d/Iron_Man_Servo

In the `config.h` file, uncomment the line `//#define SOUND` by removing the slashes. Save the file and follow the instructions for compiling and uploading the code to the board.

Connecting S.U.E. to the MK-XI Board

There is one port that needs to be plugged into S.U.E. which connects to the A.L.I.S.H.A MK-XI board. The port labeled “SUE” on the MK-XI A.L.I.S.H.A. needs to connect to the “Input” port on S.U.E.



If you are looking for 3D printable file kits for Iron Man helmet motorization, you can visit the Crash Works 3D Thingiverse page, which contains kits and documentation for Iron Man MK7, Iron Man MK46, and Iron Man MK85 helmets.

https://www.thingiverse.com/crashworks3d/designs

Thank you for your purchase, we hope you enjoy using our system and wish you the best with your project.